

Sambhav Khanna

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EDUCATION

University of Waterloo

Sep 2022 – Apr 2027

Bachelors of Computer Science (Cumulative Average 90%)

Waterloo, Canada

- Relevant Coursework: Data Structures and Algorithms, Operating Systems, Numerical Computation, Cryptography, Object-Oriented Programming, Computer Networks, Computer Architecture, Statistics

EXPERIENCE

Software Engineer Intern

Jan 2025 – Apr 2025

Geotab | C#, ASP.NET, Kubernetes, Terraform, BigQuery, Prometheus, Sentry

Waterloo, Canada

- Optimized Terraform code to reduce pod count for 50 microservices in development, cutting CPU usage by **30%**.
- Resolved a **data race** in a core service, preventing cascading failures, and implemented SQL and Redis health checks for **10** services to enable automatic pod recycling.
- Integrated Sentry for error reporting, enhancing debugging for **100+** engineers, redacted PII from GCP and Sentry logs for compliance, and created Grafana dashboards for real-time error monitoring.
- Migrated and tested services from .NET 8.0 to .NET 9.0, developed functional and unit tests to improve test coverage, and built data mappers for JSON RPC and **gRPC** signals from Tesla, Ford, and other manufacturers.

Software Engineer Intern

Jun 2024 – Sep 2024

Tallgeese AI | Python, Django, ChromaDB, Langsmith, Ollama

San Jose, United States

- Implemented accuracy evaluation pipeline to assess **agent** responses, integrating it with GitHub **CI/CD** actions.
- Developed an LRU semantic cache using **Redis** to store agent responses, reducing queries sent to LLM by **30%**.
- Enhanced language support for Chinese by resolving function-calling errors, leading to **77%** increase in accuracy.
- Resolved critical issues in document ingestion and chunking, resulting in a **96%** improvement in accuracy and a **70%** increase in user adoption, which led to **20+** new enterprise clients including **Intel**.

Software Engineer Intern

Apr 2024 – Jun 2024

Qatar Airways | Milvus, Langchain, OpenAI API, Flask, Angular.js

Doha, Qatar

- Developed a **RAG** system for storing **100+** documents, each with **200+** pages containing airplane technical data.
- Indexed the embeddings and stored them in a cloud vector database, to decrease the time to first token by **80%**.
- Implemented data cleaning pipeline, to clean tabular data, thereby increasing the accuracy of responses by **100%**.
- **Managed** an intern and assisted them in prompt engineering, RAG API development and frontend development.

Full Stack Engineer Intern

Sep 2023 – Dec 2023

Toronto Transit Commission | Python, TypeScript, React.js, Electron.js, Docker

Toronto, Canada

- Developed an encrypted password management application to store and manage **1000+** logins for **40+** databases.
- Reduced the application render time by **80%**, by migrating the codebase from Python **Tkinter** to **React.js**.
- Containerized the application, and used caching to reduce the build time from **5 minutes to 5 seconds**.
- Implemented SQL triggers and transactions to reduce data processing time, by decreasing unnecessary query runs.

PROJECTS

Predictify Pro | Python, Stripe API, Next.js, Django, GCP 🔗

- Built the website and backend for Predictify Pro, a startup, integrating a RAG chatbot and **Stripe API**.

ChatrBox | MongoDB, React.js, Express.js, Node.js, socket.io, nginx, Docker, AWS 🔗

- Built a **real-time chat** server using websocket, implementing React context to optimize **state management**.
- Containerized the app with **Docker** and deployed it on an **AWS EC2** instance behind an Nginx reverse proxy.

DocuTalk | Go, groq, Pinecone, React.js 🔗

- Developed a RAG app with Groq API and Pinecone database, achieving a **94%** success rate on table-heavy PDFs.
- Implemented **concurrent** file ingestion with, storing files in **AWS S3** for scalable processing of **100+** files.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Golang, C/C++, C#, HTML/CSS

Frameworks: Pandas, Langchain, Express.js, Next.js, Node.js, React.js, Flask, Django, gin, Tailwind CSS

Developer Tools: MongoDB, Firebase, Supabase, Postgres, MySQL, AWS, GCP, Vercel, Docker, Git, Linux, Azure